Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/14/2006
PI Name:	Shapiro, Irwin Ph.D.		
Project Title:	Test of the Equivalence Principle in an Einstein Elevator		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ishapiro@cfa.harvard.edu	**Fax:**	FY 617-495-7093
PI Organization Type:	UNIVERSITY	**Phone:**	617-495-7300
Organization Name:	Harvard University		
PI Address 1:	Smithsonian Astrophysical Observatory		
PI Address 2:	60 Garden Street, MS51		
PI Web Page:	http://cfa-www.harvard.edu/		
City:	Cambridge	**State:**	MA
Zip Code:	02138	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	03/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 03/31/2007 per PI (12/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Glashow, Sheldon (Boston University Boston MA) Iafolla, Valerio (Institute of Interplanetary Space Physics (IFSI) Rome Italy) Lorenzini, Enrico (Smithsonian Astrophysical Observatory Cambridge MA) Cheimets, Peter (Smithsonian Astrophysical Observatory Cambridge MA) Finkelstein, Nancy (Smithsonian Astrophysical Observatory Cambridge MA) Schneps, Matthew (Smithsonian Astrophysical Observatory Cambridge MA) Nozzoli, Sergio (Institute of Interplanetary Space Physics (IFSI) Rome Italy) Santoli, Francesco (Institute of Interplanetary Space Physics (IFSI) Rome Italy) Lucchesi, Davide (Institute of Interplanetary Space Physics (IFSI) Rome Italy) Cosmo, Mario (Smithsonian Astrophysical Observatory)		
Grant/Contract No.:	NAG3-2881		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics anbd unlock the mysteries of the universe obscured by gravity		

Task Description:	The goal of this grant is to carry out the experiment definition of a system for testing repeatedly the Equivalence Principle with an accuracy significantly improved with respect to the state of the art. The experimental technique utilizes a differential accelerometer (detector) in free fall inside an evacuated capsule released from a stratospheric balloon. The detector free falls for as long as 28 seconds, while rotating slowly, inside the co-moving shielding capsule which drops up to 3.8 km while providing a very high level of isolation from external noise sources. The detector measures very accurately the separation (and hence the differential rate of fall) of a pair of test masses of different composition. The experiment can be repeated periodically and, consequently, allow for fine tuning of the experiment, improvements of the detector, and testing of different materials. The ultimate objective of the experiment is to test the principle that lies at the foundation of the currently-accepted gravitational theory (general relativity), that is, the equivalence between the mass seen as a response to gravity and the mass seen as a repository of inertia. The goal of the flight experiments is to push the accuracy in testing this equivalence up to a few parts in 10^15 or, similarly, in measuring the relative acceleration in free fall of a pair of test masses of different composition (the sensing elements of the differential accelerometer) up to that accuracy. The target accuracy is about two orders of magnitude better than the present state of the art. The experiment definition activities covered by this grant are as follows: (a) assessment of science requirements and definition of engineering requirements for the experiment; (b) analysis of noise sources and their mitigation; (c) preliminary design of the free-fall system and its acceleration detector for the balloon-flight experiments; (d) construction and laboratory testing of a prototype release mechanism; (e) analysis of the theoretical implications of testing the Equivalence Principle with an improved accuracy; (f) development of techniques for extracting a small signal from data strings with dominant noise; and (g) design of the differential accelerometer with a strong rejection of noise associated with the detector's motion. In parallel to the tasks above, the group at the Institute of Space Physics (IFSI) in Rome, funded by the Italian Space Agency, develops prototype differential accelerometers and conducts key tests on the prototype accelerometers in the laboratory. The Equivalence Principle is at the foundation of gravitational theories and thus affects our understanding of the fundamental forces in nature. A negative result of the test (non violation) will impose very stringent limits on gravitational theories while a positive result (violation) will likely imply the need for a fundamental revision in the theory of general relativity and, indeed, in other basic laws developed to describe nature. The results of the tasks to be carried out under this grant will define the scientific and technical requirements of the experiment, assess its scientific/technical soundness, and lay the groundwork for the balloon experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The research activity during this grant year has focused on the following tasks: (a) analysis of the dynamics of the capsule attached to the balloon before release; and (b) definition of the shape of the sensing masses of the differential accelerometer to minimize higher-order gravitational interactions. This grant-year accomplishments are briefly summarized in the following: (1) the dynamics analysis has shown that the control system at the attachment of the capsule to the gondola (i.e., leveling mechanism) can stabilize the capsule well within the requirements for a capsule release along the vertical; and (2) a multi-objective optimization procedure was utilized to show that small modifications to the basic cylindrical shape of the proof masses strongly mitigate the effects of the higher-order gravitational mass moments produced by nearby masses (e.g., the capsule). Our non-US partners (IFSI-INAF) have carried out calibration measurements on the second differential accelerometer prototype that was recently built at the IFSI laboratory. They also tested the various elements of the measurement chain to characterize the intrinsic noise level of the prototype detector in non-cryogenic, laboratory conditions.
Bibliography Type:	Description: (Last Updated: 07/09/2007)
Articles in Peer-reviewed Journals	Shapiro, I.I.; E.C., Lorenzini; J. Ashenberg; C. Bombardelli; P.N. Cheimets; V. Iafolla; D.M. Lucchesi, S. Nozzoli; F. Santoli; and S. Glashow "Testing the Principle of Equivalence in an Einstein Elevator" Journal of Modern Physics D, (in press 2006).. , Dec-2006
Articles in Peer-reviewed Journals	Lorenzini EC, Shapiro II, Glashow S, Cheimets P, Bombardelli C, Iafolla V, Nozzoli S. "Advances in the development of the experiment to test the equivalence principle in an Einstein elevator." Advances in Space Research, (in press 2006). Corrected Proof available online 12 June 2006. , Dec-2006
Papers from Meeting Proceedings	Lorenzini EC, Shapiro II, Ashenberg J, Bombardelli C, Cheimets P, Iafolla V, Lucchesi DM, Nozzoli S, Santoli F, Glashow S. "Detector configurations for equivalence principle tests with strong separation of signal from noise." XXVIII Spanish Relativistic Meeting, E.R.E. 2005, Oviedo, Spain, Sept. 6-10, 2005. AIP Conference Proceedings 841, June 2005, p. 502-506. , Jun-2005